# Fluency Quick Check Answer Key

Teacher use · all 54 lesson checks · no student records

## Unit 2

**Lesson 2.1**

Order 7, 3, 12, 5, 9. On a line from 0 to 10 split into 5 equal jumps, what is one jump worth?

Answer: 3, 5, 7, 9, 12; each jump is 2.

**Lesson 2.2**

Which interval holds 29: 20–29 or 30–39? How many of 4, 11, 19, 21, 25 land in 10–19?

Answer: 20–29; two values (11 and 19).

**Lesson 2.3**

Put in order: 9, 4, 15, 6, 11. What number is halfway between 12 and 17?

Answer: 4, 6, 9, 11, 15; halfway is 14.5.

**Lesson 2.4**

Ordered: 3, 5, 8, 10, 12, 14, 18, 20. Find the median of the lower half.

Answer: 6.5, the average of 5 and 8.

**Lesson 2.5**

Range of 12, 7, 19, 4, 15? If Q1 = 6 and Q3 = 17, what is Q3 − Q1?

Answer: Range 15; IQR 11.

**Lesson 2.6**

Median of 4, 6, 7, 8, 40? Now take out 40 — what is the median?

Answer: 7; after removing 40, median = (6 + 7) ÷ 2 = 6.5.

**Lesson 2.7**

84 ÷ 4? 4.8 ÷ 2? 0.6 × 100?

Answer: 21; 2.4; 60.

**Lesson 2.8**

Mean of 10, 14, 8, 12, 16? What must be added to 48 to reach 60?

Answer: Mean 12; add 12 to 48.

**Lesson 2.9**

The mean is 20. How far is 26 from the mean? How far is 14?

Answer: 6 units in each case.

**Lesson 2.10**

Which measure of center fits 2, 3, 3, 4, 50 — mean or median? Why?

Answer: Median 3; the extreme value 50 pulls the mean upward.

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 3

**Lesson 3.1**

6 × 4? There are 3 dogs and 5 cats — how many animals in all?

Answer: 24; 8 animals.

**Lesson 3.2**

60 ÷ 4? If 4 pencils cost $2, what does 1 pencil cost?

Answer: 15; $0.50 per pencil.

**Lesson 3.3**

2 × 5 and 3 × 5? Complete: 2 : 3 = 8 : ___

Answer: 10 and 15; missing value 12.

**Lesson 3.4**

Start at (0, 0), move 3 right and 2 up — write the ordered pair. If each gridline is 5, where is 15?

Answer: (3, 2); 15 is the third interval from zero on a positive axis marked by fives.

**Lesson 3.5**

Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.

Answer: 12; 12 : 16.

**Lesson 3.6**

12 × 4? 1 foot = ___ inches? 2.5 m = ___ cm?

Answer: 48; 12 inches; 250 cm.

**Lesson 3.7**

4 × 1.6? 10 × 0.45? Round 6.44 to the nearest tenth.

Answer: 6.4; 4.5; 6.4.

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 4

### Lesson 4.1

37 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Answer: 37/100; 0.37; 37%.

### Lesson 4.2

0.45 as a percent? 25% as a fraction in simplest form? Which is greater, 0.6 or 55%?

Answer: 45%; 1/4; 0.6 (60%) is greater.

### Lesson 4.3

10% of 320? 1% of 320? Half of 32?

Answer: 32; 3.2; 16.

### Lesson 4.4

0.25 × 80? 3/4 of 80? Which is more: 20% of 50 or 30% of 40?

Answer: 20; 60; 30% of 40 = 12 is more than 20% of 50 = 10.

### Lesson 4.5

12 ÷ 0.25? 12 is 25% of what number? Now check: 25% of your answer.

Answer: 48; whole = 48; check 0.25 × 48 = 12.

### Use the evidence

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 5

**Lesson 5.1**

Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?

Answer: 40 $cm^2$; no, use perpendicular height (a slanted side is not generally the height).

**Lesson 5.2**

7 × 4, then half? Area of a triangle with base 7 m and height 4 m?

Answer: 28, then 14; 14 $m^2$.

**Lesson 5.3**

1/2 × (6 + 10) × 4 = ? Do the parentheses first.

Answer: 32 square units if the dimensions are lengths.

**Lesson 5.4**

A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?

Answer: 60 − 6 = 54 square units; the cutout is a rectangle, not a square.

**Lesson 5.5**

4 × 3 × 2? 1/2 × 1/3? Volume of a 4 by 3 by 2 prism?

Answer: 24; 1/6; 24 cubic units.

**Lesson 5.6**

How many faces does a rectangular prism have? How many matching pairs is that?

Answer: 6 faces in 3 matching pairs.

**Lesson 5.7**

Faces measure 12, 12, 20, 20, 15, 15 square units — what is the surface area?

Answer: 94 square units.

**Lesson 5.8**

One triangular face is 24 square inches. With four identical faces, what is the lateral area?

Answer: 96 square inches; the base is excluded from lateral area.

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 6

**Lesson 6.1**

4.8 ÷ 2? How many fourths are in 3? Reciprocal of 5?

Answer: 2.4; 12 fourths; 1/5.

**Lesson 6.2**

2 3/4 as an improper fraction? Reciprocal of 3/5? 2/3 × 5/4?

Answer: 11/4; 5/3; 5/6.

**Lesson 6.3**

$6^2$? Write 2 × 2 × 2 × 2 as a power. $10^3$?

Answer: 36; $2^4$; 1000.

**Lesson 6.4**

4 + 6 ÷ 2? 3 + $2^2$ × 5? Which operation comes first?

Answer: 7; 23. Division precedes addition in the first expression; the exponent is evaluated first in the second.

**Lesson 6.5**

Evaluate 4n + 3 when n = 5. Write an expression for "7 more than a number."

Answer: 23; n + 7 (or another defined variable plus 7).

**Lesson 6.6**

5x + 2x = ? Evaluate 2(x + 3) and 2x + 6 at x = 4 — are they the same?

Answer: 7x; both equal 14 when x = 4.

**Lesson 6.7**

All the factors of 18? First five multiples of 6? GCF of 18 and 24?

Answer: 1, 2, 3, 6, 9, 18; first five positive multiples 6, 12, 18, 24, 30; GCF 6.

**Lesson 6.8**

GCF of 18 and 24? Expand 3(x + 5). Factor 4x + 12.

Answer: 6; 3x + 15; 4(x + 3).

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 7

**Lesson 7.1**

What is the opposite of −6? Where does −3 sit compared with 0?

Answer: 6; −3 is three units left of zero.

**Lesson 7.2**

Which two integers does 2.7 fall between? Where does −11/2 go?

Answer: Between 2 and 3; −11/2 = −5.5 lies midway between −6 and −5.

**Lesson 7.3**

|−9|? How far is −4 from 0? How far from −4 to 3?

Answer: 9; 4 units; 7 units.

**Lesson 7.4**

−6 ___ −2. Order these: −3, 1.5, −1/2, 0.

Answer: <; −3, −1/2, 0, 1.5.

**Lesson 7.5**

Which quadrant holds (−4, 5)? Plot (3, −2).

Answer: Quadrant II; (3, −2) is three right and two down, in Quadrant IV.

**Lesson 7.6**

Distance from (−5, 2) to (3, 2)? What is |−5 − 3|?

Answer: 8 units; 8.

**Lesson 7.7**

Plot (1,1), (5,1), (5,4), (1,4). What is the perimeter? The area?

Answer: A 4-by-3 rectangle; perimeter 14 units; area 12 square units.

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 8

### Lesson 8.1

Is $x = 4$ a solution of $3x = 12$? Write an equation for "three times a number is 15."

Answer: Yes; $3x = 15$.

### Lesson 8.2

$m - 11 = 25$, so $m =$ ? What undoes "+ 8"?

Answer: 36; subtract 8.

### Lesson 8.3

$6x = 42$, so $x =$ ? $x \div 5 = 9$, so $x =$ ? What undoes "× 4"?

Answer: 7; 45; divide by 4.

### Lesson 8.4

Write an inequality for "you must be at least 48 inches tall." Is 50 a solution?

Answer: $h \geq 48$, where $h$ is height in inches; yes, 50 is a solution.

### Lesson 8.5

Graph $x \geq 3$. Open circle or closed? Is 3 itself a solution?

Answer: Closed circle at 3 and ray to the right; yes, 3 is included.

### Use the evidence

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Unit 9

**Lesson 9.1**

Songs cost $2 each. Which quantity is independent? Complete the table for 1, 2, and 3 songs.

Answer: Number of songs is independent; costs are $2, $4, $6.

**Lesson 9.2**

In (x, y), which value comes first? Read the point at 4 across and 12 up.

Answer: x first; (4, 12).

**Lesson 9.3**

Table: (1, 3), (2, 6), (3, 9). Write y in terms of x.

Answer: $y = 3x$.

**Lesson 9.4**

Muffins cost $3 each, so $y = 3x$. How many muffins can you buy with $24?

Answer: 8 muffins.

**Use the evidence**

Independent method accurate: continue. Arithmetic slip: brief fluency rehearsal. Model error: matching prerequisite page, then an unseen recheck. Do not use speed alone to judge conceptual understanding.

## Prerequisite source clarification

Grade 5 fraction work includes multiplication and division involving a unit fraction and a whole number. General division of fractions is Grade 6, 6.NS.A.1. Other grade labels in the original guide indicate broad prerequisite connections, not an exhaustive standards crosswalk.

References: https://www.thecorestandards.org/Math/Content/5/NF/B/7/ and https://www.thecorestandards.org/Math/Content/6/NS/A/1/